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import cv2
import numpy as np
from google.colab.patches import cv2_imshow
import urllib.request

# Download a sample image
url = "https://raw.githubusercontent.com/opencv/opencv/master/samples/data/lena.jpg"
urllib.request.urlretrieve(url, "input.jpg")

# Read the image
img = cv2.imread("input.jpg")

# Get image dimensions
rows, cols = img.shape[:2]

# Translation values
tx = 100    # Move 100 pixels to the right
ty = 50     # Move 50 pixels down

# Translation matrix
M = np.float32([[1, 0, tx],
                [0, 1, ty]])

# Perform image translation
translated = cv2.warpAffine(img, M, (cols, rows))

# Display images
print("Original Image")
cv2_imshow(img)

print("Translated Image")
cv2_imshow(translated)

# Save the output image
cv2.imwrite("translated_image.jpg", translated)
```

Original Image



Translated Image

